Pratik Pujari

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Education

University of Colorado Boulder — Boulder, CO
Master of Science in Computer Science | GPA: 3.8/4.0 — *Aug. 2025 – Present*

Sardar Patel Institute of Technology — Mumbai, India
Bachelor of Technology in Computer Engineering | GPA: 3.96/4.0 — *Oct. 2020 – May 2024*

Experience

Software Engineer Intern — May 2026 – Aug 2026
Visa Inc. — *Highlands Ranch, CO*

- Architected **VisionX**, an AI log-triage platform for ATM monitoring: **Drain3** templating, a 20-class embedding classifier (**82% F1**) with **LLM** escalation, and **HDBSCAN** clustering of ∼30K templates into ∼200 ranked issues.
- Built a **LangGraph** remediation agent that diagnoses the root cause and ships its fixes as merged **GitHub PRs**.
- Engineered a **FastAPI** + **Kafka** pipeline handling ∼**1M logs/hour** with a dead-letter queue over **PostgreSQL**.
- Cut issue triage time **50–60%** versus manual engineer review; demoed VisionX live to Visa's **VP-level** leadership.
- Wrote an **84-page** design document before building VisionX, then handed it off to a new-grad engineer on the team.
- Built a **0–100** outage **risk engine** that flags at-risk sources with leading indicators and a recommended action.

Software Developer Intern — Nov. 2023 – Dec. 2024
Aim4U Software Solutions Private Limited — *Mumbai, India*

- Architected **Dermalens**, an AI telehealth platform (discovery, booking, payments, e-prescriptions) on **Flutter** and **AWS Lambda**, integrating a **Detectron2** + ResNet skin-disease model (**90–95%** accuracy); **300+ downloads**.
- Pitched Dermalens to investors at 3+ startup competitions; shipping the judges' feedback lifted downloads by **23%**.
- Co-developed **Autobuddys**, a **Flutter** app with 5 gamified modules and **Firebase** analytics; **250+ downloads**.
- Built **GitHub Actions** CI/CD running lint, widget tests, signed builds, and **Firebase App Distribution** betas.
- Owned **Google Play** and **App Store** releases, signing, and medical-app compliance; mentored **4–5 junior interns**.

Research and Software Intern — Jan. 2023 – June 2023
Acuradyne Systems Private Limited, IIT Bombay Research Park — *Mumbai, India*

- Re-engineered the data link of an IoT radial-pulse device for aging assessment, migrating Bluetooth Classic to **BLE** on **ESP32** with chunked reassembly for **566%** faster transfer (60 to 400 points/s; 850ms to 127ms per data payload).
- Built a **Flutter** patient app with **60fps** real-time pulse charts and **1K+** daily **PDF** clinical reports via a Node.js API.
- Added **BLE** auto-connect and disconnect with power optimization, extending device battery life between sessions.
- Shipped **OTA** firmware over **BLE** with checksum-verified transfer and rollback, cutting production time ∼**40%**.

Projects

Echo – Real-time Chat App | *Kotlin, Jetpack Compose, Hilt, Room* | Android | Backend — Mar. 2025 – Sept. 2025

- Architected the **Kotlin/Compose** client: **11 Gradle modules**, **MVI**, Hilt, and a **50+**-component design system.
- Engineered a **WebSocket** client with heartbeats, jittered backoff, and receipts for a **10K+**-connection backend.
- Wrote **4** custom **Lint** rules (theming, hardcoded strings, memory leaks, security) plus **JUnit**/MockK/**Espresso** tests.
- Implemented **FCM** notifications with inline actions, encrypted local storage, and offline-first sync to a Go backend.

JetScan – Document Scanner | *Kotlin, Compose, CameraX, OpenCV* | GitHub | Demo — May 2024 – Nov. 2024

- Built a **35K+ line** open-source **Kotlin** scanner in **Jetpack Compose** (MVVM) with **CameraX** and **OpenCV**.
- Engineered **100ms** angle-invariant document detection using LAB color space, **Canny**, **HoughLinesP**, and contours.
- Modeled **Room** DAO relations, added perspective correction and **PDF**/image export, and **Document AI** OCR.
- Covered camera and scanning flows with **JUnit**, MockK, and **Espresso** tests in a **GitHub Actions** release pipeline.

Skycast – Weather App | *Kotlin, Compose, Material 3, Retrofit* | GitHub — Aug. 2025

- Built a **Material 3** weather app with 10-day forecasts, hourly air quality, and weather news from **4** API providers.
- Structured clean architecture with a **BaseViewModel**(State, Event, Action) and a typed **Retrofit** Result adapter.
- Added GPS location, city search, saved cities in **Room**, and day/night theming with **Material You** dynamic color.
- Built **Navigation Compose** flows with **Lottie** animations, **Coil** image loading, and persisted unit preferences.

Technical Skills

Languages: Kotlin (Coroutines/Flow), Java, Dart, Python, Go, SQL
Android & UI: Jetpack Compose, Material 3, Navigation Compose, CameraX, Room, Retrofit/OkHttp, Flutter
Architecture: MVVM/MVI, Clean Architecture, Multi-module Gradle, Hilt/Dagger, Offline-first Sync
Testing & Release: JUnit, MockK, Espresso, Lint, Android Profiler, R8, GitHub Actions, Play Console, Git